

Innovation & Design Programme

 NUS
National University of Singapore

College of Design and Engineering

CDE2301 Value Creation in Innovation

Jovan Tan
Lecturer, Engineering Design and Innovation Centre
jovan@nus.edu.sg

Session 1 (13 Aug 2026)
Course Introduction & Understanding Value Creation

© Jovan Tan 2026. All Rights Reserved.

NUS Restricted

1

Innovation & Design Programme

 NUS
National University of Singapore

College of Design and Engineering

Your Instructor



Dr. Jovan Tan
陈俊文 博士

Director
Sustainability Action & Innovation Lab (SAIL@NUS)

Adjunct Faculty
Singapore-MIT Alliance for Research and Technology (SMART)
NUS Distinguished Senior Fellowship Program

Lecturer
Engineering Design and Innovation Centre (EDIC)

Resident Fellow
Residential College 4 (RC4)

Email: jovan@nus.edu.sg
Website: <https://cde.nus.edu.sg/edic/staff/jovan-tan/>
LinkedIn: <https://www.linkedin.com/in/jovantan/>

© Jovan Tan 2026. All Rights Reserved.

NUS Restricted

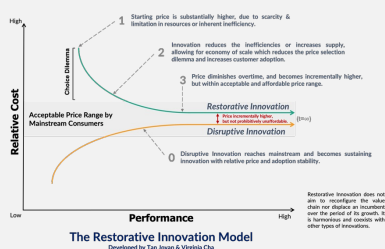
2

Your Instructor



Teaching and Research Interests

1. **Innovations**, Strategies and Leadership for **Sustainable Development**
2. **Education** for **Sustainable Development**



LASER FRAMEWORK

TO DESIGN INNOVATION FOR THE CIRCULAR ECONOMY

1. Alignment of Internal & External Factors
2. Life Cycle of the Product Innovation
3. System Enablers & Value Chains
4. Economic Viability
5. Resource Planning for Efficient Execution
6. Alignment for Commitment & Action

This work is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

<https://www.weforum.org/agenda/2022/08/what-is-restorative-innovation-and-why-does-it-matter/>

<https://www.weforum.org/agenda/2023/03/designing-innovations-circular-economy/>

© Jovan Tan 2026. All Rights Reserved.

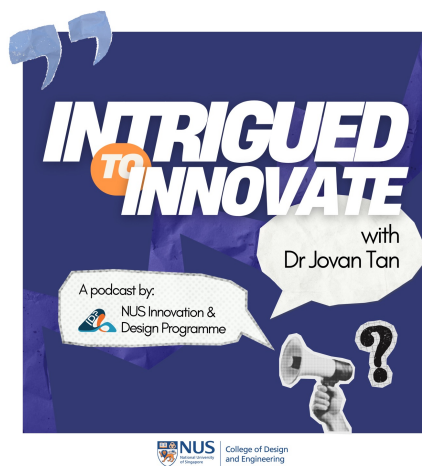
3



4

Podcast

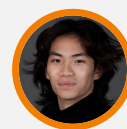
Innovation & Design Programme

College of Design
and Engineering

- Showcases **inspiring stories of young innovators** tackling complex, real-world challenges through creative, interdisciplinary solutions.
- Shares **valuable lessons, insights, and perspectives** gained from their innovation journeys.
- Occasionally features **thought leaders** discussing topics in innovation, design, and entrepreneurship.



Jovan Tan
Creator &
Host



Low Tse Han
Executive
Producer

© Jovan Tan 2025. All Rights Reserved.

5

NUS Restricted

Your Instructor

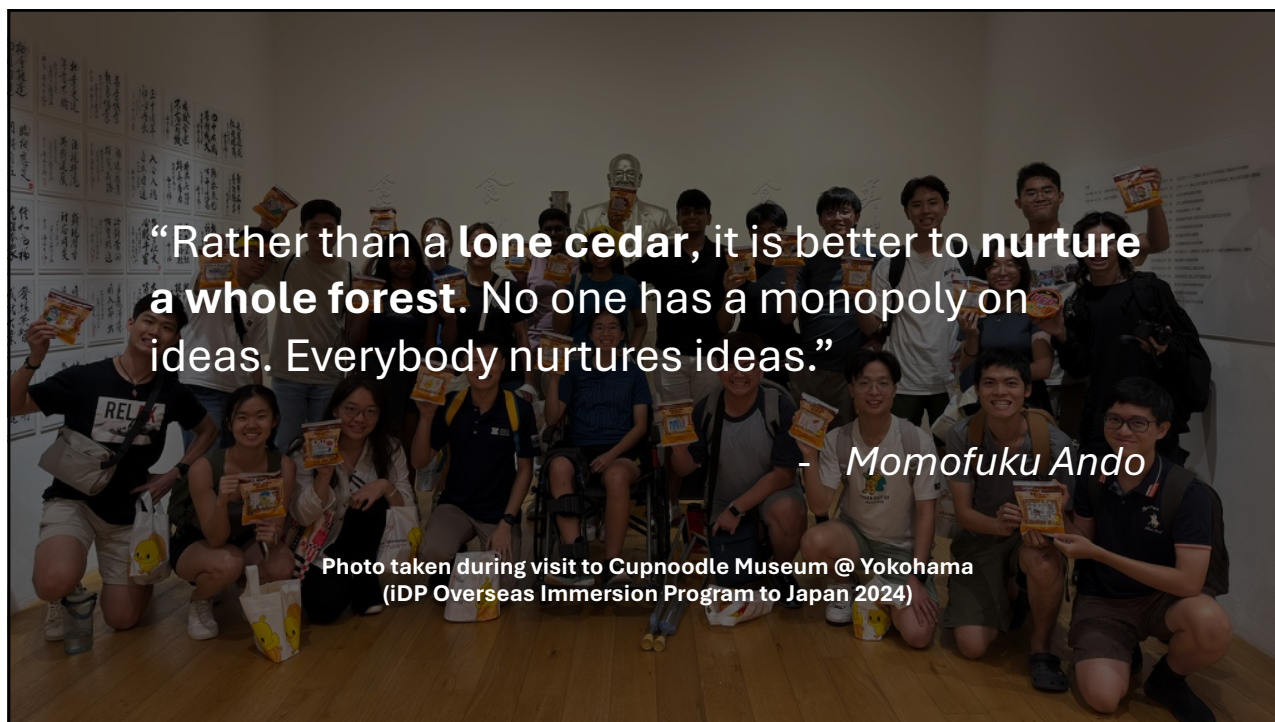
Innovation & Design Programme

College of Design
and Engineering

© Jovan Tan 2026. All Rights Reserved.

6

NUS Restricted

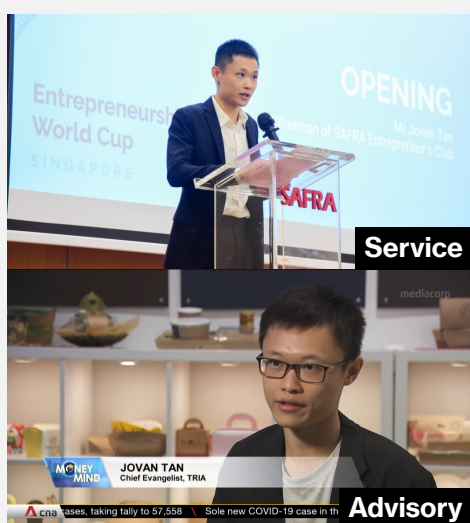


Your Instructor

Innovation & Design Programme



College of Design and Engineering

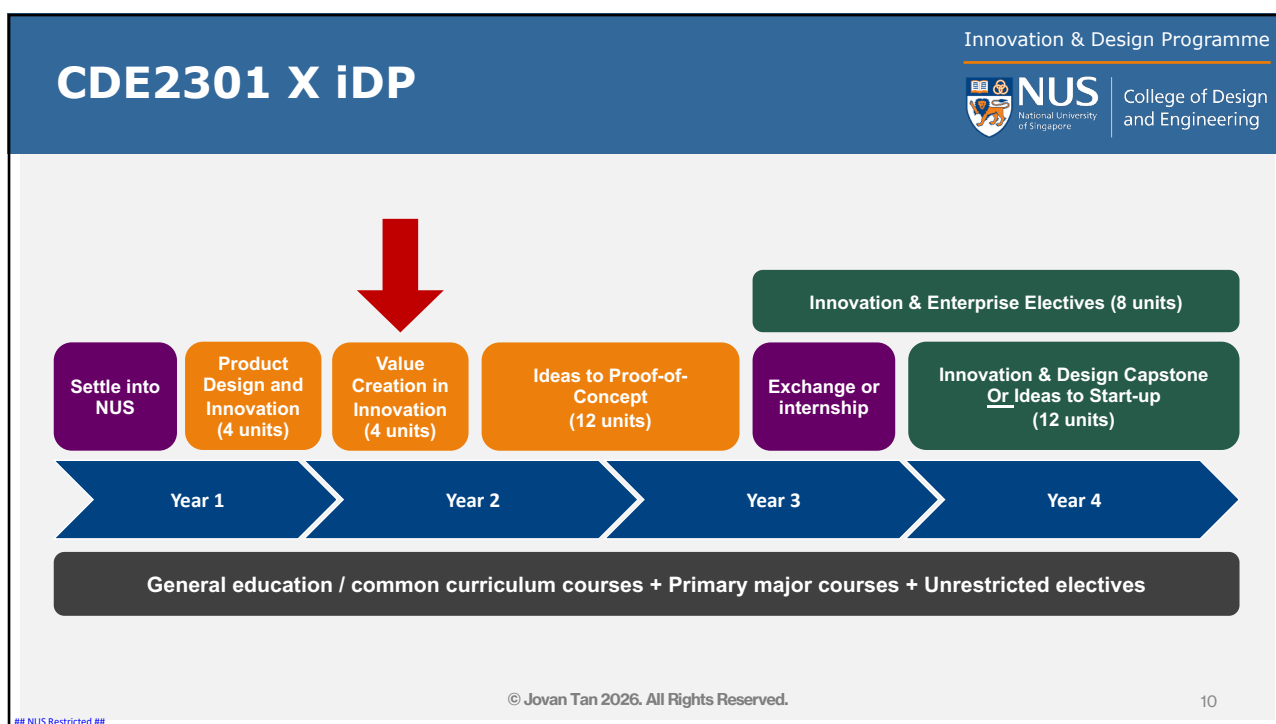


Affiliations

- Fellow, World Economic Forum (2022-2024)
- Editorial Board Member, Materials Circular Economy (Springer Nature)
- Committee Member, Zero Waste Taskforce, NUS
- Advisor, Ucaneco GmbH
- Certified Lead Verifier, ISO 14064 GHG Accounting and Reporting
- Immediate Past Chairman & Advisor, SAFRA Entrepreneurs' Club

© Jovan Tan 2026. All Rights Reserved.

8



CDE2301 X iDP

Innovation & Design Programme

National University of Singapore

College of Design and Engineering

INNOVATION happens here!

Source: Dr Elliot Law, iDP Program Coordinator

“Put simply, it (design thinking) is a discipline that uses the designer’s sensibility and methods to **match people’s needs** with what is **technologically feasible** and **what a viable business strategy can convert into customer value and market opportunity.**”

Tim Brown, IDEO

On Design Thinking, in Harvard Business Review, June 2008

NUS Restricted
© Jovan Tan 2026. All Rights Reserved.
11

CDE2301 X iDP

Innovation & Design Programme

National University of Singapore

College of Design and Engineering

INNOVATION happens here!

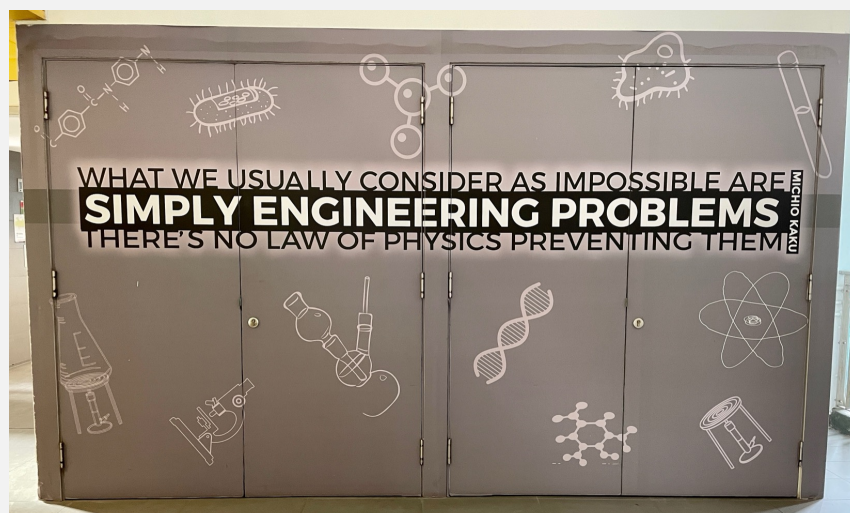
Source: Dr Elliot Law, iDP Program Coordinator

CDE3301

NUS Restricted
© Jovan Tan 2026. All Rights Reserved.
12

CDE2301 X iDP

Innovation & Design Programme

College of Design
and Engineering

© Jovan Tan 2026. All Rights Reserved.

13

NUS Restricted

CDE2301 X iDP

Innovation & Design Programme

College of Design
and Engineering**PET Bottle-to-Bottle Recycling**

(Desirable, Feasible, but not widely adopted. Why?)



© Jovan Tan 2026. All Rights Reserved.

Source: Plastics Recycling
Association of Singapore

14

NUS Restricted

CDE2301 X iDP

Innovation & Design Programme

College of Design
and Engineering

Food Waste > Black Soldier Fly Larvae > Animal Feed

(Desirable, Feasible, but not widely adopted. Why?)



Source: NUS News

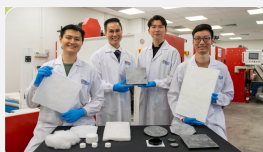
© Jovan Tan 2026. All Rights Reserved.

15

NUS Restricted

CDE2301 X iDP

Innovation & Design Programme

College of Design
and Engineering

Scientists from the NUS Department of Mechanical Engineering have developed innovative aerogels for radiative cooling and electromagnetic wave absorption.

Using plastic waste, the team engineered thin-film aerogels that function as thermal insulators and radiative coolers, which can be applied to the roofs of buildings to reduce indoor temperatures. The team also created aerogels that effectively absorb electromagnetic energy, shielding both humans and sensitive equipment in our increasingly digital world.

Read more: <https://nus.edu/3VrEGXa>

@nusnews



Researchers from the NUS Institute for Health Innovation & Technology (iHealthtech) have discovered a new way to stimulate muscle cells to produce and release proteins with anticancer properties by using brief and mild electromagnetic pulses of particular characteristics.

This drug-free, non-invasive approach could open the doors for new developments in cancer therapy and cancer biomarker discovery.

<https://nus.edu/3UOutCG>

@nusnews

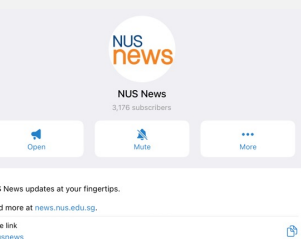


Wireless technologies like Wi-Fi, Bluetooth, and 5G rely on radio frequency signals to send and receive data.

In a breakthrough for green energy, a team led by NUS scientists has developed a new prototype of an energy harvesting module that can now convert ambient or 'waste' radio frequency signals into direct current voltage that can be used to power small electronic devices without the use of batteries.

<https://nus.edu/3YALQmf>

@nusnews



© Jovan Tan 2026. All Rights Reserved.

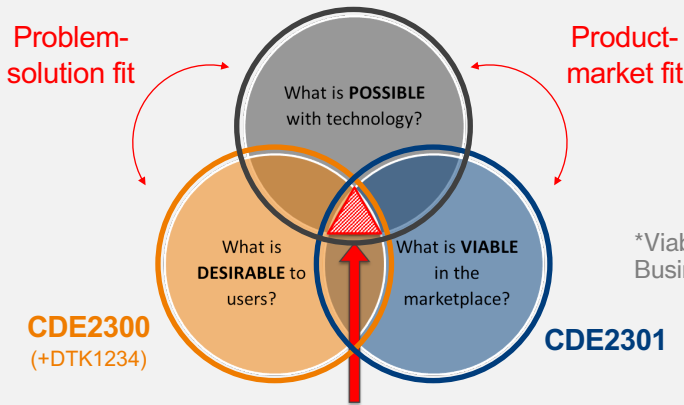
16

NUS Restricted

CDE2301 X iDP

Innovation & Design Programme
 NUS
 National University of Singapore
 College of Design and Engineering

CDE3301



CDE2300 (+DTK1234) **CDE2301**

INNOVATION happens here!

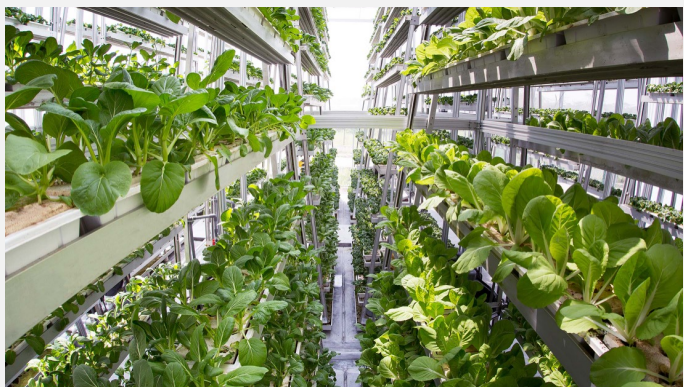
*Viable = Sustainable Business Opportunity

NUS Restricted
© Jovan Tan 2026. All Rights Reserved.
Source: Dr Elliot Law, iDP Program Coordinator 17

CDE2301 X iDP

Innovation & Design Programme
 NUS
 National University of Singapore
 College of Design and Engineering

Vertical Farming in Singapore
 (Desirable, Feasible, Adopted, but is it **Viable**?)



Source: Skygreens

NUS Restricted
© Jovan Tan 2026. All Rights Reserved.
18

CDE2301 X iDP

Innovation & Design Programme

College of Design
and Engineering

Reverse Vending Machines

(Desirable, Feasible, Adopted, but is it **Viable**?)Source: Dr Amy Khor
Facebook Page

© Jovan Tan 2026. All Rights Reserved.

19

NUS Restricted

CDE2301 X iDP

Innovation & Design Programme

College of Design
and Engineering

Reverse Vending Machines

(Desirable, Feasible, Adopted, but is it **Viable**?)

Thank you for your positive support for the Recycle N Save initiative.

In preparation for the launch of the Beverage Container Return Scheme,
**all the rewards currently offered under Recycle N Save
 will be discontinued after 31 July 2023.**

We look forward to your continued support to recycle more and recycle right, to do your part for the
 environment!

Source: recyclensave.sg

© Jovan Tan 2026. All Rights Reserved.

20

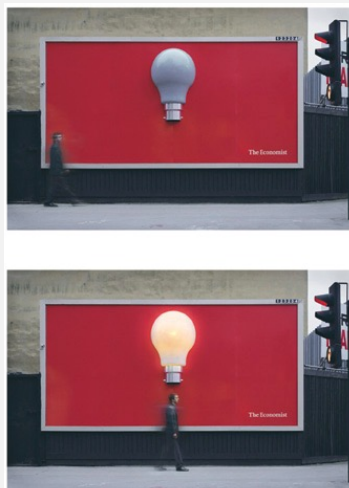
NUS Restricted

CDE2301 X iDP

Innovation & Design Programme



College of Design and Engineering



Ideas vs Opportunities

- Ideas are “a dime a dozen.”
- Opportunities are business ideas that offer the **potential** for a return on invested capital that more than offsets the costs of that capital.

© Jovan Tan 2026. All Rights Reserved.

Source: Prof Virginia Cha

21

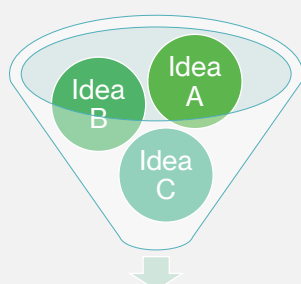
NUS Restricted

Long Road to Success

Innovation & Design Programme



College of Design and Engineering



Opportunities



Opportunity Assessment

(Technological Feasibility, Viability, etc.)

- Customer Discovery & Validation
- Value Proposition
- Industry Structure
- Market Sizing
- Competitive Strategies


**Business Plan
& Execution**

© Jovan Tan 2026. All Rights Reserved.

Source: Prof Virginia Cha

22

NUS Restricted

Innovation & Design Programme

Business Plan

NUS National University of Singapore College of Design and Engineering

An effective business plan needs to answer the following top 10 questions:

The diagram features a central sphere labeled "Effective Business Plan". Ten lines radiate from this sphere to ten questions arranged in two columns. The questions are:

- What is your current situation?
- What is your goal?
- How will you reach this goal?
- What do you offer?
- How big is the business you are entering?
- Who are the key players?
- How will you differentiate yourself?
- What will be your marketing plan?
- What are the economics of your business?
- What is the capital requirement to get your business started?

© Jovan Tan 2026. All Rights Reserved. Source: Prof Virginia Cha 23

NUS Restricted

Innovation & Design Programme

Business Plan

NUS National University of Singapore College of Design and Engineering

An effective business plan needs to answer the following top 10 questions:

The diagram features a central sphere labeled "Effective Business Plan". Four lines radiate from this sphere to four questions arranged in a single column. The questions are:

- What is your current situation?
- What is your goal?
- How will you reach this goal?
- What do you offer?

Weeks 2 & 3:

- Value Proposition

© Jovan Tan 2026. All Rights Reserved. Source: Prof Virginia Cha 24

NUS Restricted

Business Plan

Innovation & Design Programme

National University of Singapore

College of Design and Engineering

An effective business plan needs to answer the following top 10 questions:

- Who are the key players?
- How will you differentiate yourself?

Week 4:

- Industry Structure
- Market Sizing
- Competitive Strategies

NUS Restricted

© Jovan Tan 2026. All Rights Reserved.

Source: Prof Virginia Cha

25

Business Intelligence (Week 5)

Innovation & Design Programme

National University of Singapore

College of Design and Engineering

NUS Libraries

Instructors from NUS Libraries will conduct an activity-based learning workshop on locating market, consumer, and company information using NUS Libraries' subscribed databases.

Content Coverage:

- Market Research via Frost & Sullivan
- Consumer Research via Passport
- Company Research via D&B Hoovers
- Mainstream Media Research
- IEEE Citation Style vs APA Citation Style
- Awareness of Zotero
- Mapping Business Model Canvas to subscribed databases.

NUS Restricted

© Jovan Tan 2026. All Rights Reserved.

26

Business Plan

Innovation & Design Programme
 **NUS**
 National University of Singapore
 College of Design and Engineering

An effective business plan needs to answer the following top 10 questions:

Weeks 9 & 10:

- **Business Model Design**



Effective Business Plan

- **What are the economics of your business?**

NUS Restricted
© Jovan Tan 2026. All Rights Reserved.
Source: Prof Virginia Cha
27

Business Plan

Innovation & Design Programme
 **NUS**
 National University of Singapore
 College of Design and Engineering

An effective business plan needs to answer the following top 10 questions:

Week 11:

- **Go-to-Market Strategy**



Effective Business Plan

- **What will be your marketing plan?**

NUS Restricted
© Jovan Tan 2026. All Rights Reserved.
Source: Prof Virginia Cha
28

Business Plan

Innovation & Design Programme

National University of Singapore

College of Design and Engineering

An effective business plan needs to answer the following top 10 questions:

```

graph TD
    A((Effective Business Plan)) --- B[What is your current situation?]
    A --- C[What is your goal?]
    A --- D[How will you reach this goal?]
    A --- E[What do you offer?]
    A --- F[How big is the business you are entering?]
    A --- G[Who are the key players?]
    A --- H[How will you differentiate yourself?]
    A --- I[What will be your marketing plan?]
    A --- J[What are the economics of your business?]
    A --- K[What is the capital requirement to get your business started?]
  
```

NUS Restricted

© Jovan Tan 2026. All Rights Reserved.

Source: Prof Virginia Cha

29

Strategic Narrative

Innovation & Design Programme

National University of Singapore

College of Design and Engineering

Week 12: Put It All Together & Develop Your Pitch

NUS Restricted

© Jovan Tan 2026. All Rights Reserved.

Source: Prof Virginia Cha

30

Learning Outcomes

Innovation & Design Programme

College of Design
and Engineering

At the end of this course, you should be able to:

- Understand the **industry value chain** and **competitive rivalry** within the industry.
- Propose new product ideas and develop **value propositions** for these product ideas.
- Explore the viability of a product idea using the **business model canvas**.
- Understand and devise **competitive strategies** for a product idea.
- Understand **go-to-market approaches** and create a product idea's initial market launch plan.
- Conduct **secondary research** using relevant literature to support a proposed product idea.

© Jovan Tan 2026. All Rights Reserved.

31

NUS Restricted

Schedule

Innovation & Design Programme

College of Design
and Engineering

Date	Week	Topics
13 Aug 2026	1	Course Introduction & Understanding Value Creation
20 Aug 2026	2	Designing a Value Proposition for all Stakeholders I
27 Aug 2026	3	Designing a Value Proposition for all Stakeholders II
03 Sep 2026	4	Understanding Industry Structure and Competitive Strategies
10 Sep 2026	5	Business Intelligence Workshop by NUS Libraries
17 Sep 2026	6	Mid-Point Presentation
Recess		
01 Oct 2026	7	No Class (Consult with TAs)
08 Oct 2026	8	No Class (Replace with CDE Innovation Day on 21 Oct 2026)
15 Oct 2026	9	Crafting Your Business Model with Business Model Canvas I
22 Oct 2026	10	Crafting Your Business Model with Business Model Canvas II
29 Oct 2026	11	Go-To-Market Strategies
05 Nov 2026	12	Pitching Your Innovation to Customers
12 Nov 2026		Final Presentation

** Schedule is subject to change. Always refer to Canvas for the latest schedule.

© Jovan Tan 2026. All Rights Reserved.

32

NUS Restricted

Session Structure

Innovation & Design Programme

 NUS
National University of Singapore

College of Design and Engineering

1st Hour (6pm – 7pm):
Report Backs

- Report back on previous week's assignments.
- **Report backs are not graded.**
- Report backs provide you with a safe learning environment for you to:
 - **discuss** your learnings.
 - hone your **communication skills** and business/market orientation skills.
 - receive **regular feedback** regarding your progress.

NUS Restricted

© Jovan Tan 2026. All Rights Reserved.

33

Session Structure

Innovation & Design Programme

 NUS
National University of Singapore

College of Design and Engineering

1st Hour (6pm – 7pm):
Report Backs

↓

2nd Hour (7pm – 8pm):
Learning & Discussions

↓

3rd Hour (8pm – 9pm):
Workshop

- Learn new concepts and frameworks
- Classroom discussions
- Time allocated for you to work on your assignments and seek clarification with the course team.

NUS Restricted

© Jovan Tan 2026. All Rights Reserved.

34

Deliverables and Gradings	
Innovation & Design Programme  NUS National University of Singapore College of Design and Engineering	
Assessment Components	Weightage
Team Assessment [50%]	
Group Project <i>Students will work in groups to critically identify, evaluate and articulate the value drivers of their respective projects using tools and frameworks taught in class.</i>	40%
Strategic Narrative <i>Students will pitch the value proposition of their projects to stakeholders.</i>	10%
Individual Assessment [50%]	
Class Participation <i>Students will be awarded points for attendance and active participation in the classroom discussion.</i>	10%
Course Reflection <i>Students will reflect on their learnings after each session. Students also will deeply and holistically reflect on their (a) CDE2301 learning journey, (b) the good and the bad of this module, and (c) key takeaways. Guiding questions will be provided.</i>	30%
Peer Evaluation <i>Students will evaluate their group mate's performance for the group project. Template will be provided.</i>	10%
© Jovan Tan 2026. All Rights Reserved. ## NUS Restricted ##	
35	

Class Participation	
Innovation & Design Programme  NUS National University of Singapore College of Design and Engineering	
- Points are awarded to learners who participate and contribute to the classroom discussion by asking quality questions, providing constructive comments, or other appropriate means (excluding participation in their own report-back sessions). - Our Tutors will record participation, and learners will be ranked and grouped into 10 bands at the end of the semester. Based on their bands, points will be awarded.	
© Jovan Tan 2026. All Rights Reserved. ## NUS Restricted ##	
36	

Course Reflections

Innovation & Design Programme

College of Design
and Engineering

1. Session Reflections (20%)

- Our post-session reflection follows the **3-2-1 active and reflective learning** strategy proposed by Zygoris-Coe, V., Wiggins, M.B. and Smith, L.H.
- 1 point will be given for submitting an entry.
- An extra point will be given for entries demonstrating deep reflection and coherent articulation of their thoughts.
- A maximum of 20 points will be awarded for post-session reflections.

2. Course Reflections (10%)

- Deeply and holistically reflect on your (a) CDE2301 learning journey, (b) the good and the bad of this module, and (c) key takeaways.
- Guiding questions will be provided.

© Jovan Tan 2026. All Rights Reserved.

37

NUS Restricted

Activity

Innovation & Design Programme

College of Design
and Engineering

Breakout Session: Team Formation & Project Selection

Coordinator: Yuan Yun & Yu Xuan

© Jovan Tan 2026. All Rights Reserved.

38

NUS Restricted

Report Back for Week 2

Innovation & Design Programme

College of Design
and Engineering

1. **Create your name tents.**
2. **Form your groups (5 per group) and inform our TAs.**
3. **Identify a *desirable* and *feasible* business opportunity (via NUS News)**
 - **Validate your identified opportunity with our TAs** before proceeding with the rest of the assignment.
 - **Report back:**
 - Introduce the team (team name, members, etc.)
 - What is your identified business opportunity, and why did you select it?
 - What are the products and services?
 - What will likely be the path to viability?

Reminder: You will work on this opportunity with your group for the entire course.

© Jovan Tan 2026. All Rights Reserved.

39

NUS Restricted